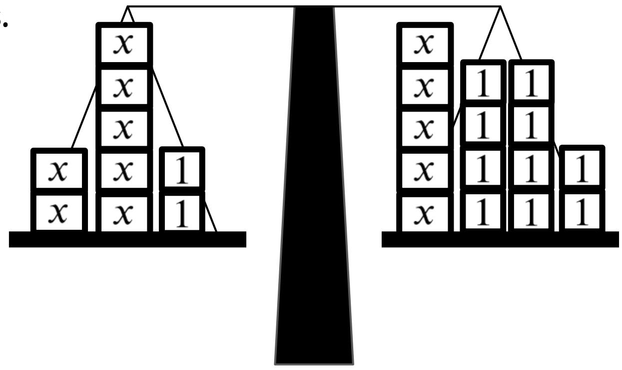
Rearranging to solve linear equations

Task A: Representing equations with unknowns on both sides

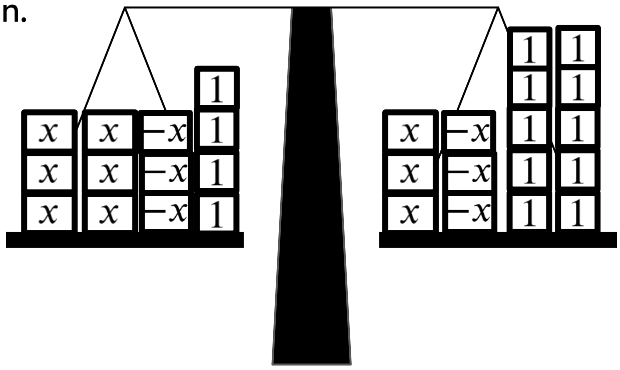
1) Build an equation with unknowns on both sides.



Starting with this balance showing $2x = 8$ show what you need to add to both sides to make the equation $7x + 2 = 5x + 10$

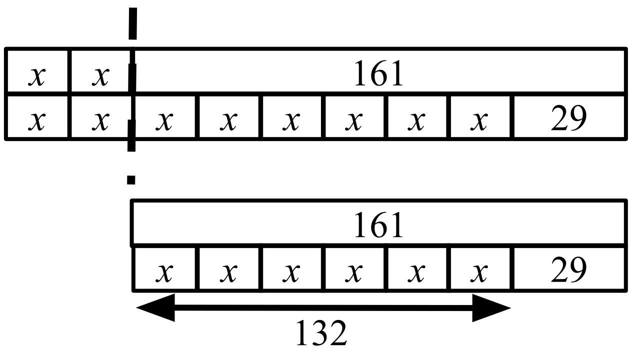
$$\begin{aligned} 2x &= 8 \\ 2x + (5x) &= 8 + (5x) \\ 7x &= 5x + 8 \\ 7x + (2) &= 5x + 8 + (2) \\ 7x + 2 &= 5x + 10 \end{aligned}$$

2) 'Un-do' this balance model to solve this equation.



$$\begin{aligned} 6x + 4 &= 3x + 10 \\ 6x + 4 + (-3x) &= 3x + 4 + (-3x) \\ 3x + 4 &= 10 \\ 3x + 4 + (-4) &= 10 + (-4) \\ 3x &= 6 \\ x &= 2 \end{aligned}$$

3) Use this bar model to solve this equation.



Write your algebraic notation alongside.

$$\begin{aligned} 2x + 161 &= 8x + 29 \\ 161 &= 6x + 29 \\ 132 &= 6x \\ x &= 22 \end{aligned}$$

Task B: Solving equations with unknowns on both sides

1) Solve this equation in two different ways:

a)

$$\begin{aligned} 3x + 2 &= 7x - 10 \\ 3x + 2 + (-3x) &= 7x - 10 + (-3x) \\ 2 &= 4x - 10 \\ 2 + (10) &= 4x - 10 + (10) \\ 12 &= 4x \\ x &= 3 \end{aligned}$$

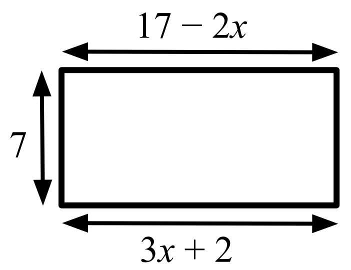
$$\begin{aligned} 3x + 2 &= 7x - 10 \\ 3x + 2 + (10) &= 7x - 10 + (10) \\ 3x + 12 &= 7x \\ 3x + 12 + (-3x) &= 7x + (-3x) \\ 12 &= 4x \\ x &= 3 \end{aligned}$$

b)

$$\begin{aligned} 2x + 2 &= 23 - 5x \\ 2x + 2 + (5x) &= 23 - 5x + (5x) \\ 7x + 2 &= 23 \\ 7x + 2 + (-2) &= 23 + (-2) \\ 7x &= 21 \\ x &= 3 \end{aligned}$$

$$\begin{aligned} 2x + 2 &= 23 - 5x \\ 2x + 2 + (-2) &= 23 - 5x + (-2) \\ 2x &= 21 - 5x \\ 2x + (5x) &= 21 - 5x + (5x) \\ 7x &= 21 \\ x &= 3 \end{aligned}$$

2) Find the perimeter of this rectangle.



$$\begin{aligned} 17 - 2x &= 3x + 2 \\ 17 &= 5x + 2 \\ 15 &= 5x \\ x &= 3 \end{aligned}$$

Perimeter $= 2(11 + 7) = 2(18) = 36$

3) Solve: $\frac{1}{8}x - 7 = 3 - 7x$

$$8\left(\frac{1}{8}x - 7\right) = 8\,(3 - 7x)$$

$$\begin{aligned} x - 56 &= 24 - 56x \\ 57x - 56 &= 24 \\ 57x &= 80 \end{aligned}$$

$$x = \frac{80}{57}$$